

Generation Interconnection System Impact Study Report

For Queue Project Q16-0638

Ryceville 138 kV

May 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0638. The Interconnection Customer proposes a 52.8 MW solar photovoltaic facility interconnecting at the Ryceville 138 kV substation in the Maryland area. The request was received on June 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 326 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$105,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Holland Cliff – Chalk Point 138 KV is overloaded to 245.0% of its 237 MVA emergency rating for the loss of the University Of Northern Iowa – Stranger Creek 138 KV. The project contributes 5.1% DFAX (2.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Holland Cliff – Chalk Point 138 KV	Rebuild line with higher-capacity conductor	\$4,922,000	2% (\$105,000)
	Total allocated to project		\$105,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.